

This book provides you with practice examples of how programming concepts are applied, with a collection of over 80 games, simulations, and digital art programs. These aren't code snippets; they're full, runnable Python programs. You can copy their code to become familiar with how they work, experiment with your own changes, and then attempt to re-create them on your own as practice. After a while, you'll start to get ideas for your own programs and, more importantly, know how to go about creating them.

How to Design Small Programs

Programming has proven to be a powerful skill, creating billion-dollar tech companies and amazing technological advances. It's easy to want to aim high with your own software creations, but biting off more than you can chew can leave you with half-finished programs and frustration. However, you don't need to be a computer genius to code fun and creative programs.

The Python programs in this book follow several design principles to aid new programmers in understanding their source code:

Small Most of these programs are limited to 256 lines of code and are often significantly shorter. This size limit makes them easier to comprehend. The choice of 256 is arbitrary, but 256 is also 2^8 , and powers of 2 are lucky programmer numbers.

Text based Text is simpler than graphics. Since the source code and program output are both text, it's easy to trace the cause and effect between `print('Thanks for playing!')` in the code and `Thanks for playing!` appearing on the screen.

No installation needed Each program is self-contained in a single Python source file with the `.py` file extension, like `tictactoe.py`. You don't need to run an installer program, and you can easily post these programs online to share with others.

Numerous There are 81 programs in this book. Between board games, card games, digital artwork, simulations, mathematical puzzles, mazes, and humor programs, you're bound to find many things you'll love.

Simple The programs have been written to be easy to understand by beginners. Whenever I had to choose between writing code using sophisticated, high-performance algorithms or writing plain, straightforward code, I've chosen the latter every time.

The text-based programs may seem old school, but this style of programming cuts out the distractions and potholes that downloading graphics, installing additional libraries, and managing project folders bring. Instead, you can just focus on the code.

Who Is This Book For?

This book is written for two groups of people. The people in the first group are those who have already learned the basics of Python and programming